

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: MLA0301

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3,BL4,BL5)

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

Design and Develop Model

Value-Equivalence Prediction and Policy Optimization

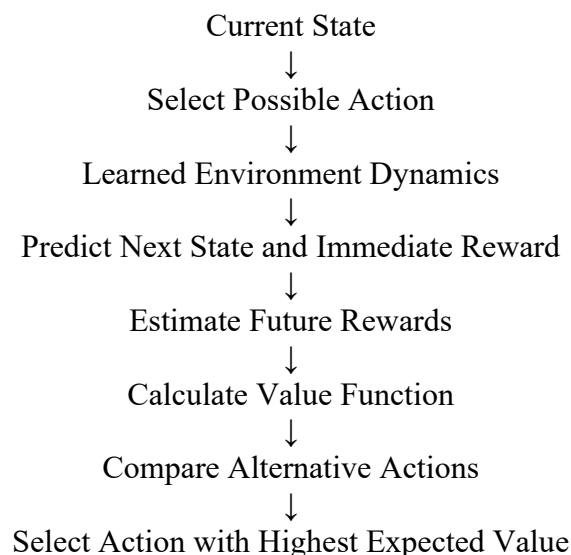
21. Value-Equivalence Prediction for Estimating Future Rewards

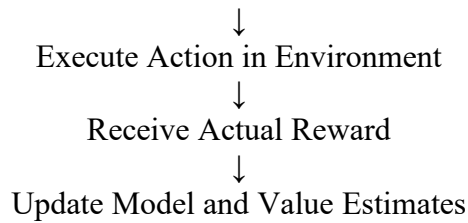
Answer:

Value-equivalence prediction is a Model-Based Reinforcement Learning technique used to estimate the **future value or expected cumulative reward** of an action without requiring the agent to experience the complete future sequence in the real environment.

The process begins with the agent observing the **current state**. It considers a possible action and uses the learned environment model to predict the next state and reward. The predicted information is then used by the value function to estimate the future return.

Detailed flow:



**Key Idea:**

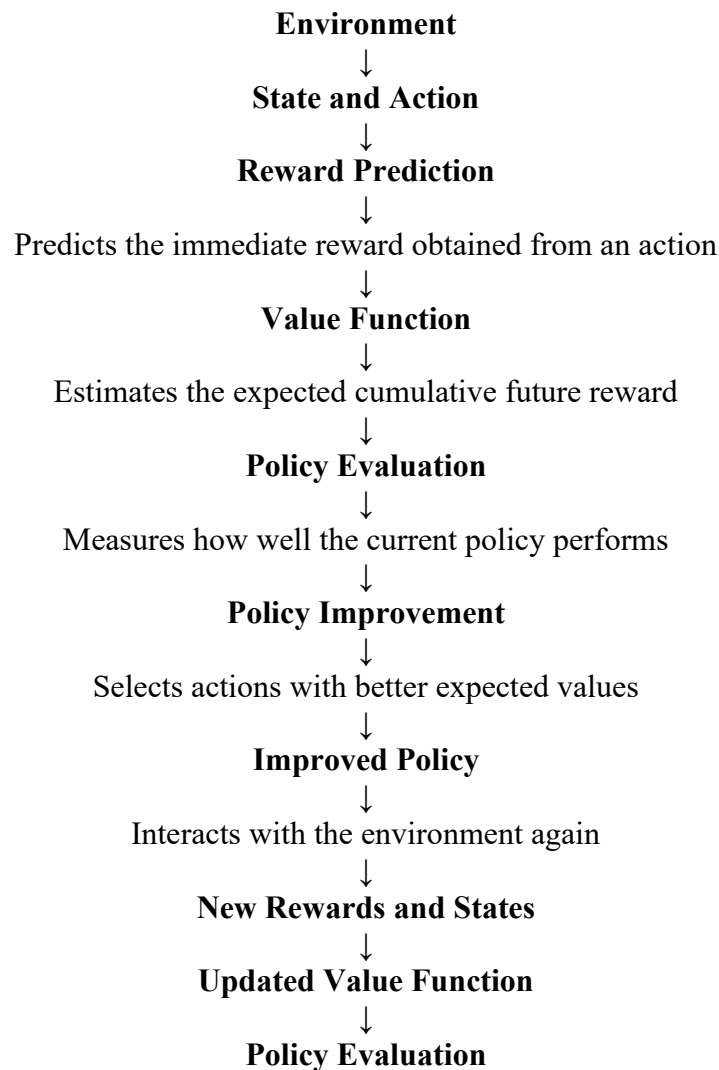
The important objective is not necessarily to reproduce every detail of the environment. Instead, the model should preserve the information required to make **accurate value and reward predictions**.

For example, in a warehouse robot, the model can predict whether moving toward a particular location will result in a shorter route and higher reward. The agent uses this prediction to select a better action.

22. Concept Map: Value Functions, Reward Prediction, Policy Evaluation and Policy Improvement

Answer:

The four concepts are closely connected because they form a continuous cycle of **evaluating and improving an RL policy**.

Concept Map:

↓
Policy Improvement

Relationship Between the Concepts:**Reward Prediction → Value Function**

Reward prediction estimates the immediate or future reward associated with an action. The value function uses these rewards to estimate the total expected return.

Value Function → Policy Evaluation

The value function helps determine how well the current policy performs in different states.

Policy Evaluation → Policy Improvement

After evaluating the current policy, the agent identifies actions that provide higher expected rewards.

Policy Improvement → New Policy

The policy is updated to prefer better actions.

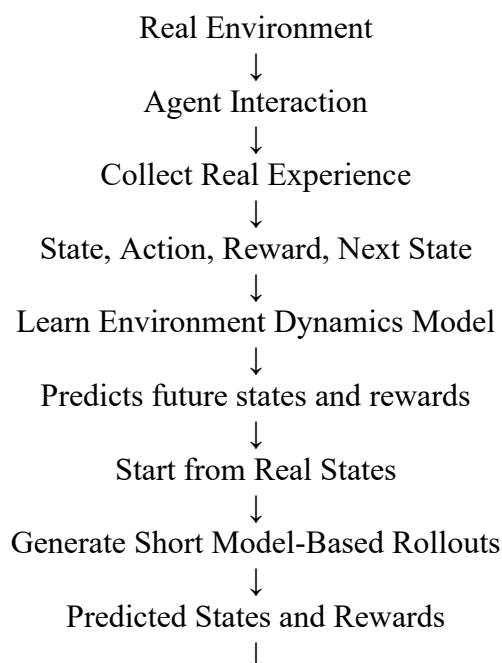
Overall Relationship:**Reward Prediction**

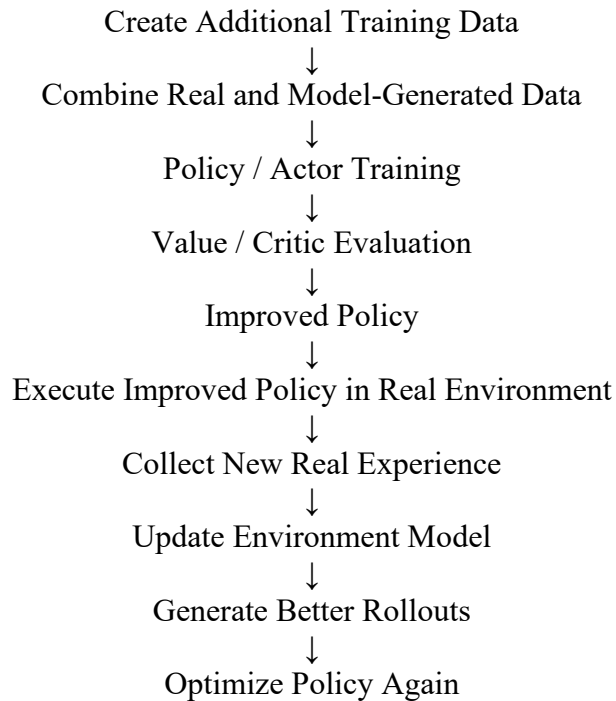
- provides reward information
- **Value Function**
- evaluates expected return
- **Policy Evaluation**
- measures current performance
- **Policy Improvement**
- creates a better policy
- **Environment Interaction**
- generates new experience
- repeats the process.

This repeated cycle allows the RL agent to gradually converge toward an **optimal policy**.

23. Complete Model-Based Policy Optimization (MBPO) Framework**Answer:**

Model-Based Policy Optimization (MBPO) combines **real environmental experience, learned environment dynamics, simulated data generation, and policy optimization.**

Complete Framework:

**Main Components:****1. Real Environment:**

Provides reliable real-world transitions and rewards.

2. Environment Model:

Learns the dynamics of the environment from real data.

3. Model-Based Rollouts:

The learned model generates short simulated trajectories.

4. Policy Optimization:

The agent uses real and simulated experiences to improve its policy.

5. Iterative Learning:

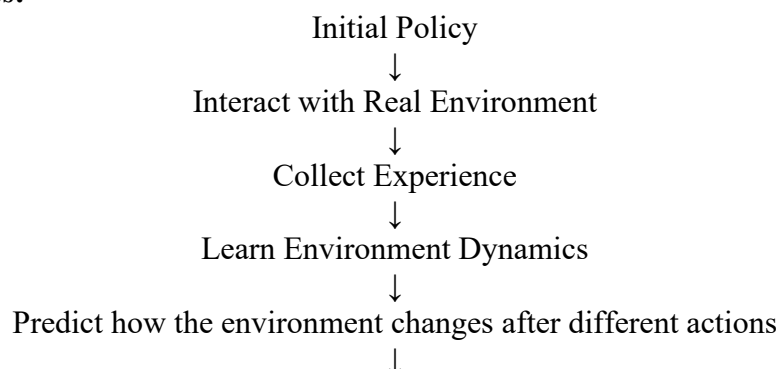
New real experience continuously improves both the model and policy.

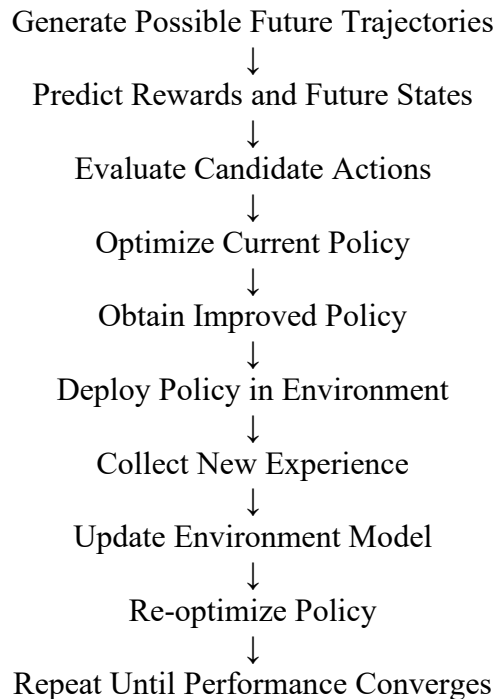
Important Feature:

MBPO generally uses **short model-generated rollouts** because long rollouts can accumulate prediction errors. This provides additional training data while reducing the effect of model inaccuracies.

24. Iterative Policy Optimization Using Learned Environment Dynamics**Answer:**

Iterative policy optimization allows an RL agent to repeatedly improve its policy using a learned model of the environment.

Detailed Process:

**Explanation:**

Initially, the agent has a policy that may not be optimal. It interacts with the environment and collects transition data. This data is used to learn the environment dynamics.

The learned model predicts the outcomes of possible actions. The agent uses these predictions to evaluate different strategies and improve its policy.

After applying the improved policy to the real environment, new experience is collected. The environment model is updated again, and policy optimization is repeated.

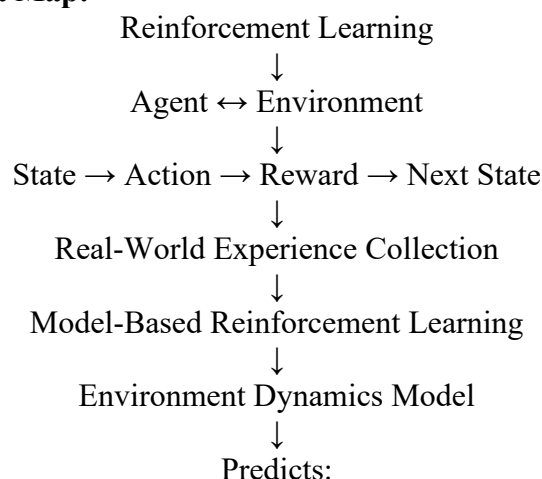
Main Objective:

Better Environment Model → Better Predictions → Better Policy → Better Actions → Higher Reward

The process continues until the policy reaches satisfactory performance or convergence.

25. Comprehensive Concept Map: Unified Reinforcement Learning Framework**Answer:**

The major Model-Based RL techniques can be integrated into a single framework where **environment modelling, planning, prediction, data generation, and policy optimization** support one another.

Comprehensive Concept Map:

Next States + Rewards + Future Outcomes

From here, the framework connects to several major techniques:

1. Analytic Gradient Computation

Environment Model

- Differentiable Model
- Calculate Gradients
- Determine Direction of Policy Improvement
- Update Policy Parameters

Analytic gradients allow the system to directly calculate how changes in actions or policy parameters affect future rewards.

2. Sampling-Based Planning

Environment Model

- Generate Multiple Possible Actions
- Sample Future Trajectories
- Predict Outcomes
- Evaluate Trajectories
- Select Best Action

Sampling-based planning is useful when the system cannot easily calculate an exact optimal solution.

3. Model-Based Data Generation

Learned Environment Model

- Generate Simulated Experiences
- State
- Action
- Predicted Reward
- Predicted Next State
- Additional Training Data

This reduces the need for expensive real-world interactions.

4. Value-Equivalence Prediction

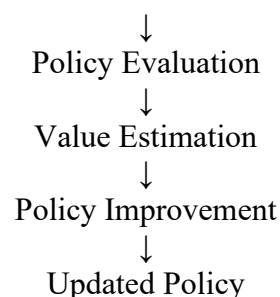
Predicted States and Rewards

- Estimate Future Return
- Value Function
- Compare Actions
- Identify High-Value Actions

The focus is on preserving the information necessary for **accurate value and decision prediction**.

5. Model-Based Policy Optimization

Real Experience + Model-Generated Experience



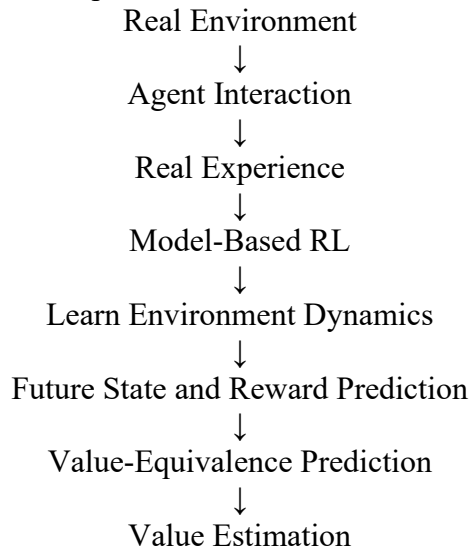


Real Environment Interaction

The improved policy generates new real-world experience, which is again used to improve the environment model.

Unified Relationship

The complete relationship can be represented as:



At the same time:

Learned Environment Model

- **Sampling-Based Planning**
- Future Trajectory Generation
- Action Evaluation

And:

Learned Environment Model

- **Model-Based Data Generation**
- Simulated Experiences
- Additional Training Data

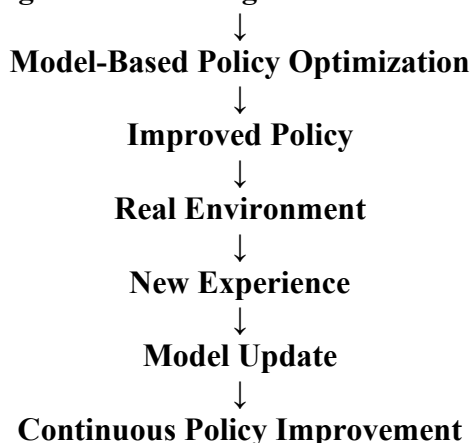
And:

Differentiable Environment Model

- **Analytic Gradient Computation**
- Gradient-Based Policy Improvement

Finally:

Value Prediction + Sampling-Based Planning + Model-Based Data + Analytic Gradients



Final Unified Concept:

Model-Based RL acts as the main framework, while **Analytic Gradient Computation, Sampling-Based Planning, Model-Based Data Generation, and Value-Equivalence Prediction** provide different methods for predicting and improving decisions. These techniques support **Model-Based Policy Optimization**, which continuously improves the agent's policy using both real and model-generated experience.

Overall cycle:

Environment → Experience → Model Learning → Prediction/Planning → Policy Optimization → Improved Action → Environment → Continuous Learning

Code of Conduct:

*I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness